

KIRILLOVKA, Ukraine

WMO: 346090

Lat: 47.3297N

Lon: 36.3350E

Elev: 221

StdP: 98.70

Time Zone: 2.00 (EUE)

Period: 86-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-14.5	-19.4	0.7	-16.3	-17.0	0.9	-14.1	17.4	-3.0	16.6	-2.2	1.6	50	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	31.4	20.4	29.8	19.8	28.1	19.3	21.7	28.6	20.9	27.5	20.1	26.5	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.7	24.6	18.6	13.8	23.8	17.7	13.0	23.2	64.6	28.5	61.4	27.3	58.5	26.6	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	14.8	11.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-3.0	-2.3	2.8	9.1	14.8	19.0	21.0	20.9	15.9	9.2	2.2	-2.1		
	DBStd	9.94	5.46	5.51	4.52	4.17	3.84	2.99	3.02	3.18	4.31	4.93	5.66	5.02		
	HDD10.0	1738	402	345	227	66	7	0	0	0	4	76	237	374		
	HDD18.3	3668	660	578	483	278	121	28	9	13	96	285	485	632		
	CDD10.0	1382	0	0	3	38	156	271	342	337	183	50	3	0		
	CDD18.3	270	0	0	0	0	12	49	93	91	24	1	0	0		
	CDH23.3	2511	0	0	0	1	98	383	898	921	207	3	0	0		
	CDH26.7	737	0	0	0	0	8	77	279	314	59	0	0	0		
(14) Wind	WSAvg	3.0	3.4	3.1	3.8	3.4	3.0	2.9	2.6	2.3	2.5	3.8	2.7	3.1		
(15) Precipitation	PrecAvg	563	44	36	43	46	44	71	55	49	39	36	45	46		
	PrecMax	952	152	81	89	105	103	211	150	162	143	86	124	120		
	PrecMin	375	20	4	19	0	0	19	1	3	0	5	0	8		
	PrecStd	134	27	21	20	28	27	46	45	47	33	20	31	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	12.3	18.0	22.8	27.7	30.4	33.0	33.8	31.8	23.8	14.8	11.1		
		MCWB	6.4	9.3	10.1	13.8	17.9	19.9	22.0	20.1	19.6	15.5	12.1	10.6		
	2%	DB	5.9	7.5	14.6	20.3	25.8	28.6	31.2	31.7	28.2	21.2	12.9	7.3		
		MCWB	5.4	6.2	9.1	12.8	17.1	19.5	20.8	20.0	18.2	14.5	11.1	6.5		
	5%	DB	4.5	5.6	11.7	18.3	24.3	27.1	29.6	29.9	25.8	18.5	11.3	5.3		
		MCWB	4.0	4.8	7.7	11.4	16.6	19.5	20.2	19.7	17.1	13.9	9.9	4.8		
	10%	DB	3.1	4.1	9.2	16.5	22.1	25.5	27.8	28.1	23.3	16.2	9.4	3.9		
		MCWB	2.7	3.3	6.2	10.8	15.3	18.6	19.3	19.4	16.0	12.8	8.3	3.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	9.7	11.2	14.6	20.8	22.0	23.0	23.1	21.0	17.2	13.2	10.7		
		MCDB	7.1	11.8	16.7	20.6	26.3	27.1	30.0	28.5	29.6	21.0	14.2	11.2		
	2%	WB	5.5	6.6	9.5	13.5	18.6	20.9	21.8	21.6	19.1	15.9	11.5	6.5		
		MCDB	5.9	7.3	13.9	18.4	23.7	26.3	29.3	28.8	26.2	18.5	12.7	7.1		
	5%	WB	3.9	5.0	8.1	12.3	17.3	20.1	20.9	20.6	17.8	14.6	10.1	4.9		
		MCDB	4.3	5.7	11.4	17.2	22.0	25.1	27.9	27.7	23.9	17.6	11.0	5.2		
	10%	WB	2.7	3.5	6.6	11.2	16.1	19.3	19.9	19.7	16.7	12.9	8.5	3.4		
		MCDB	3.1	4.1	8.7	15.6	20.5	24.2	26.3	26.8	21.8	16.5	9.4	3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	6.2	7.5	9.7	10.5	9.9	11.6	12.2	10.5	9.1	6.0	4.5		
		MCDBR	4.6	6.9	11.7	13.2	12.3	12.4	13.3	13.7	13.5	11.8	7.5	4.6		
	5% WB	MCWBR	4.4	5.8	7.2	7.2	5.9	5.5	5.5	6.1	6.4	6.6	5.7	4.3		
		MCWBR	4.5	6.8	10.5	11.1	10.0	10.7	12.1	13.0	12.4	9.5	6.0	4.5		
(40) Clear-Sky Solar Irradiance	taub		0.330	0.360	0.390	0.444	0.409	0.410	0.434	0.432	0.404	0.371	0.347	0.332		
	taud		2.257	2.198	2.205	2.089	2.281	2.301	2.238	2.234	2.289	2.350	2.342	2.300		
	Ebn at Noon		726	783	820	807	857	855	827	811	799	767	706	677		
	Edh at Noon		80	105	121	150	128	126	133	128	111	90	74	69		
(44) All-Sky Solar Radiation	RadAvg		0.98	1.89	2.99	4.63	6.01	6.33	6.32	5.71	4.06	2.43	1.17	0.76		
	RadStd		0.18	0.33	0.37	0.44	0.52	0.47	0.35	0.37	0.35	0.33	0.18	0.12		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.76	N/A	N/A	+1.18	N/A	N/A	-140	-186	+126	+85

Nomenclature: See separate page